



**FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY  
(FOEBEIT)**

**BCL1244 GAME PROGRAMMING**

**SEMESTER**

**MAY-AUGUST 2026**

**CASE STUDY 20%**

**(INDIVIDUAL)**

**BACHELOR IN COMPUTER SCIENCE-ODL**

**Declaration**

I hereby confirm that I have read and understood the terms and conditions of the assessment plagiarism policy. I declare that this case study is entirely my own work and has been completed without any form of plagiarism. All sources of information, if any, have been properly acknowledged.

| No | Student Full Name | Student ID | Signature |
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**Task Overview**

You are required to analyse the given game scenario and propose improvements based on game design principles and Unity implementation concepts. Your answer must be clear, structured, and supported with explanations.

**QUESTION 1****“Cyber City Rescue Mission” Scenario:**

A 2D action-platform game called Cyber City Rescue Mission is being developed in Unity. The player controls a hero who must rescue civilians while avoiding enemies and environmental hazards in a futuristic city.

However, during early development, the design team notices:

- Enemy behaviour is too predictable and reduces challenge
- Scoring system does not motivate replay ability
- Game UI does not clearly communicate player health and objectives
- Level progression is not clearly structured

You are assigned to redesign the game system focusing on game design principles and player experience.

**(a) Gameplay System Analysis**

Identify **THREE (3)** design problems in the current system and explain why they reduce game quality. **(8 marks)**

**(b) Game Design Enhancement**

Propose **THREE (3)** improvements to enhance:

- challenge and difficulty curve
- reward and scoring system
- player motivation and engagement

**(9 marks)**

**(c) Unity System Integration**

Explain how your improvements can be supported using Unity. Provide **THREE (3)** relevant Unity implementations. **(8 marks)**

**-END OF QUESTIONS-**